How long will my food delivery take?

Predicting delivery time with machine learning, and a live app

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LaTeX

Start

Why predict delivery time?

- Customers want an **honest ETA**
- Restaurants and dispatchers need it to **plan couriers**
- **Goal:** predict delivery time in minutes from the order, the route and the courier
- A **regression** problem. Target: `Time_taken (min)`

Final deliverable

A Streamlit app anyone can use to get an estimate, with a range.

Jump to: ▸ Data ▸ Model ▸ Results ▸ Live demo

The data: Zomato food delivery orders

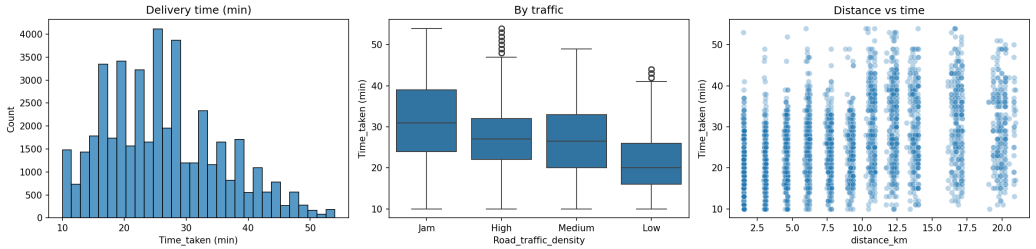
- Kaggle: Zomato food delivery dataset. Each row is one order: route, weather, traffic, vehicle and courier
- Delivery times range from **10 to 54 minutes**
- The data comes from Indian cities



Cleaning: what we fixed and why

Problem in the data	What we did
Padded text, `"NaN "`	Stripped spaces, made real blanks
Mixed date and time formats	Auto-detected; built weekday, weekend, hour
Only raw coordinates	Computed **distance in km**
Codes that look like numbers	Kept as **categories**
Pickup time	**Removed:** unknown at order time (leakage)
Rows with 4+ blanks	Dropped (1.5%); others filled in the pipeline

What the data shows

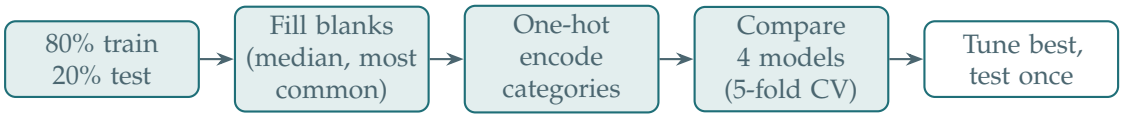


Most orders take 15 to 30 min. The longest are over 50.

Traffic matters. Median is about 31 min in a jam and about 20 min in light traffic.

Distance alone is weak. Times vary widely at every distance, so we need several features together.

How the model was built



Steps 2 and 3 sit in one pipeline, learned on training data only

Cross-validation error (average miss, minutes). Lower is better.



The gap between Ridge and the tree models shows the relationships are not linear.

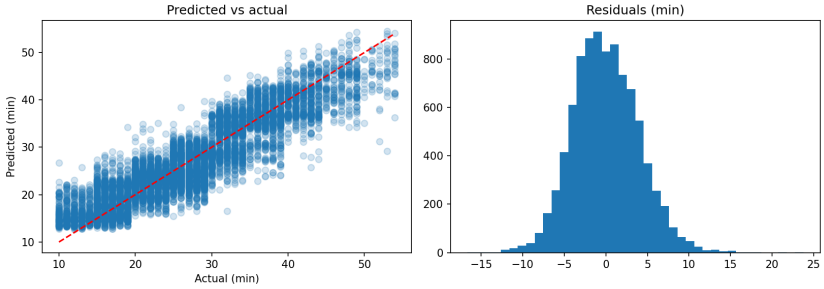
Results on 8,978 orders the model never saw

	Guess the average	Our model
Average error (MAE)	7.59 min	**3.16 min**
RMSE	9.39 min	**3.97 min**
R^2	0.00	**0.82**
Orders within ±5 min	38%	**81%**

Typically off by about **3 minutes**. Cross-validation error (3.15) matches the test error, so the model is **not overfit**.

▸ Where do the errors fall?

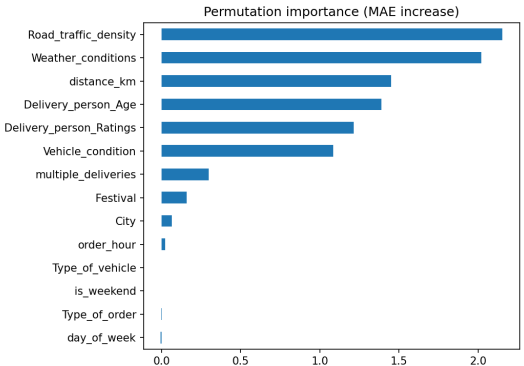
Where the errors fall



Left: predicted vs actual (red line = perfect). Right: errors are centred on 0 and mostly within ±5 min.

Very fast and very slow orders are pulled toward the average.

What drives delivery time



- **Traffic** and **weather** matter most
- Then **distance**, courier **age**, **rating** and vehicle condition
- Weekday, hour and order type add almost nothing

Importance = how much the error grows when one column is shuffled. It shows what the model uses, not what causes delays.

Try it live

https://YOUR-APP-NAME.streamlit.app

Three things to try

1. Press **Quiet sunny day**, then **Rush hour in a storm**, and watch the estimate move
2. Open **What changes the time** to see one input at a time
3. Save two orders and **compare** them side by side

▸ Backup screenshot

Takeaways and limits

What we learned

- Typical error **about 3 min**; **81%** of orders within 5 min
- Traffic, weather, distance and the courier drive the time
- The model is packaged as a **live app**

Limits and next steps

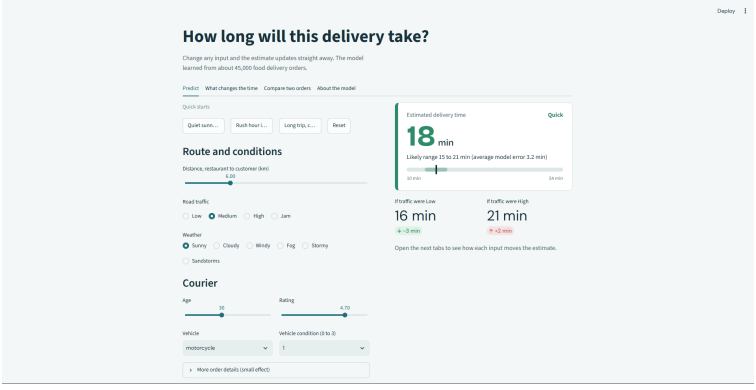
- Indian-city data: retrain on **local data** for Phnom Penh
- Random split, not tested over time
- Courier details are unknown to a customer, so it suits **dispatchers**
- Next: live traffic, time-based testing

Thank you!

Try the app:

https://YOUR-APP-NAME.streamlit.app

Backup: the app



Back to demo